Experiment 18: First-Order Predicate Calculus Parser in Python

1. Aim

To write a Python program to implement a simple First-Order Predicate Calculus (FOPC) parser for basic logical expressions.

2. Procedure

1. Import `Expression` from the `nltk.sem` module.
2. Initialize the parser function `Expression.fromstring`.
3. Define basic logical expressions using quantifiers, predicates, and operators.
4. Parse the logical strings into FOPC expression objects.
5. Display the parsed logical expressions.

3. Program

```
from nltk.sem import Expression

read_expr = Expression.fromstring

e1 = read_expr("man(socrates)")
print("Parsed Expression 1:", e1)

e2 = read_expr("all x.(man(x) -> mortal(x))")
print("Parsed Expression 2:", e2)
```

4. Output

```
Parsed Expression 1: man(socrates)
Parsed Expression 2: all x.(man(x) -> mortal(x))
```

5. Result

Thus, the Python program was successfully executed to implement a simple FOPC parser for logical expressions using `Expression.fromstring`.